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Programiz
C++ Online Compiler

main.cpp

```
1 #include <iostream>
2 #include <fstream>
3 #include <string>
4
5 using namespace std;
6
7 class Book {
8 private:
9     int bookID;
10    string title;
11    string author;
12    bool issued;
13
14 public:
15    // Add book details
16    void addBook() {
17        cout << "Enter Book ID: ";
18        cin >> bookID;
19        cin.ignore();
20        cout << "Enter Book Title: ";
21        getline(cin, title);
22        cout << "Enter Author Name: ";
23        getline(cin, author);
24        issued = false;
25        saveBook();
26        cout << "Book Added Successfully!\n";
27    }
28
29    // Save book to file
30    void saveBook() {
31        ofstream file("library.txt", ios::app);
32        file << bookID << endl;
33        file << title << endl;
34        file << author << endl;
35        file << issued << endl;
36    }
37
38    // Display all books
39    static void displayAllBooks() {
40        ifstream file("library.txt");
41        if (!file.is_open()) {
42            cout << "Library is empty!\n";
43            return;
44        }
45        string line;
46        while (getline(file, line)) {
47            cout << line << endl;
48        }
49        file.close();
50    }
51
52    // Search book
53    void searchBook() {
54        string searchText;
55        cout << "Enter Search Text: ";
56        getline(cin, searchText);
57        ifstream file("library.txt");
58        if (!file.is_open()) {
59            cout << "Library is empty!\n";
60            return;
61        }
62        string line;
63        while (getline(file, line)) {
64            if (line.find(searchText) != string::npos) {
65                cout << line << endl;
66            }
67        }
68        file.close();
69    }
70
71    // Issue book
72    void issueBook() {
73        int id;
74        cout << "Enter Book ID: ";
75        cin >> id;
76        cin.ignore();
77        ifstream file("library.txt");
78        if (!file.is_open()) {
79            cout << "Library is empty!\n";
80            return;
81        }
82        string line;
83        while (getline(file, line)) {
84            if (line.find(to_string(id)) != string::npos) {
85                cout << line << endl;
86                issued = true;
87                saveBook();
88                cout << "Book Issued Successfully!\n";
89            }
90        }
91        file.close();
92    }
93
94    // Return book
95    void returnBook() {
96        int id;
97        cout << "Enter Book ID: ";
98        cin >> id;
99        cin.ignore();
100        ifstream file("library.txt");
101        if (!file.is_open()) {
102            cout << "Library is empty!\n";
103            return;
104        }
105        string line;
106        while (getline(file, line)) {
107            if (line.find(to_string(id)) != string::npos) {
108                cout << line << endl;
109                issued = false;
110                saveBook();
111                cout << "Book Returned Successfully!\n";
112            }
113        }
114        file.close();
115    }
116
117    // Update book
118    void updateBook() {
119        int id;
120        cout << "Enter Book ID: ";
121        cin >> id;
122        cin.ignore();
123        ifstream file("library.txt");
124        if (!file.is_open()) {
125            cout << "Library is empty!\n";
126            return;
127        }
128        string line;
129        while (getline(file, line)) {
130            if (line.find(to_string(id)) != string::npos) {
131                cout << line << endl;
132                string title;
133                string author;
134                bool issued;
135                cout << "Enter Book Title: ";
136                getline(cin, title);
137                cout << "Enter Author Name: ";
138                getline(cin, author);
139                issued = false;
140                saveBook();
141                cout << "Book Updated Successfully!\n";
142            }
143        }
144        file.close();
145    }
146
147    // Remove book
148    void removeBook() {
149        int id;
150        cout << "Enter Book ID: ";
151        cin >> id;
152        cin.ignore();
153        ifstream file("library.txt");
154        if (!file.is_open()) {
155            cout << "Library is empty!\n";
156            return;
157        }
158        string line;
159        while (getline(file, line)) {
160            if (line.find(to_string(id)) != string::npos) {
161                cout << line << endl;
162                remove("library.txt");
163                cout << "Book Removed Successfully!\n";
164            }
165        }
166        file.close();
167    }
168
169    // Main menu
170    void mainMenu() {
171        int choice;
172        do {
173            cout << "LIBRARY MANAGEMENT SYSTEM\n";
174            cout << "1. Add Book\n";
175            cout << "2. Display All Books\n";
176            cout << "3. Search Book\n";
177            cout << "4. Issue Book\n";
178            cout << "5. Return Book\n";
179            cout << "6. Exit\n";
180            cout << "Enter Your Choice: ";
181            cin >> choice;
182            cin.ignore();
183            switch (choice) {
184                case 1: addBook(); break;
185                case 2: displayAllBooks(); break;
186                case 3: searchBook(); break;
187                case 4: issueBook(); break;
188                case 5: returnBook(); break;
189                case 6: exit(0); break;
190            }
191        } while (choice != 6);
192    }
193
194 int main() {
195    Book b;
196    b.mainMenu();
197    return 0;
198 }
```

Output

```
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 1
Enter Book ID: 56
Enter Book Title: Krishna avadharam
Enter Author Name: Krishna Chaitanya
Book Added Successfully!
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 2
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 3
Search By:
1. Title
2. Author
Enter Choice: 1
Enter Search Text: shivan books
```

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main.cpp

```
114
115 }
116
117 // Update file
118 void updateFile() {
119     ifstream file("library.txt");
120     ofstream temp("temp.txt");
121     int id;
122     string t;
123     bool isIssued;
124     while (file >> id) {
125         file.ignore();
126         getline(file, t);
127         temp << t << endl;
128         if (id == bookID) {
129             temp << bookID << endl;
130             temp << title << endl;
131             temp << author << endl;
132             temp << issued << endl;
133         } else {
134             temp << id << endl;
135             temp << t << endl;
136             temp << a << endl;
137             temp << isIssued << endl;
138         }
139     }
140     file.close();
141     temp.close();
142     remove("library.txt");
143     rename("temp.txt", "library.txt");
144 }
145
146 // Display all books
147 static void displayAllBooks() {
148     ifstream file("library.txt");
149     if (!file.is_open()) {
150         cout << "Library is empty!\n";
151         return;
152     }
153     string line;
154     while (getline(file, line)) {
155         cout << line << endl;
156     }
157     file.close();
158 }
```

Output

```
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 1
Enter Book ID: 56
Enter Book Title: Krishna avadharam
Enter Author Name: Krishna Chaitanya
Book Added Successfully!
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 2
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 3
Search By:
1. Title
2. Author
Enter Choice: 1
Enter Search Text: shivan books
```


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main.cpp

```
223 if (!issued)
224     cout << "\nStatus : Issued\n";
225 else
226     cout << "\nStatus : Available\n";
227
228     found = true;
229 }
230 }
231
232 if (!found)
233     cout << "\nBook Not Found\n";
234
235     file.close();
236 }
237 }
238
239 int main() {
240     Book book;
241
242     int choice, id;
243
244     while (true) {
245
246         cout << "\n===== LIBRARY MANAGEMENT SYSTEM =====\n";
247         cout << "1. Add Book\n";
248         cout << "2. Display All Books\n";
249         cout << "3. Search Book\n";
250         cout << "4. Issue Book\n";
251         cout << "5. Return Book\n";
252         cout << "6. Exit\n";
253
254         cout << "\nEnter Your Choice: ";
255         cin >> choice;
256
257         switch (choice) {
258             case 1:
259                 book.addBook();
260                 break;
261             case 2:
262                 book.displayAllBooks();
263                 break;
264             case 3:
265                 book.searchBook();
266                 break;
267             case 4:
268                 book.issueBook();
269                 break;
270             case 5:
271                 book.returnBook();
272                 break;
273             case 6:
274                 break;
275             default:
276                 cout << "\nInvalid Choice\n";
277                 continue;
278         }
279     }
280 }
```

Output

```
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 1
Enter Book ID: 56
Enter Book Title: Krishna avadharam
Enter Author Name: Krishna Chaitanya
Book Added Successfully!
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 2
===== BOOK LIST =====
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 3
Search By:
1. Title
2. Author
Enter Choice: 1
Enter Search Text: shivan books
```

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main.cpp

```
258 case 1:
259     book.addBook();
260     break;
261 case 2:
262     book.displayAllBooks();
263     break;
264 case 3:
265     book.searchBook();
266     break;
267 case 4:
268     cout << "\nEnter Book ID: ";
269     cin >> id;
270     if (book.loadBook(id))
271         book.issueBook();
272     else
273         cout << "\nBook Not Found\n";
274     break;
275 case 5:
276     cout << "\nEnter Book ID: ";
277     cin >> id;
278     if (book.loadBook(id))
279         book.returnBook();
280     else
281         cout << "\nBook Not Found\n";
282     break;
283 case 6:
284     cout << "\nThank You\n";
285     return 0;
286 default:
287     cout << "\nInvalid Choice\n";
288     continue;
289 }
```

Output

```
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 1
Enter Book ID: 56
Enter Book Title: Krishna avadharam
Enter Author Name: Krishna Chaitanya
Book Added Successfully!
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 2
===== BOOK LIST =====
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Your Choice: 3
Search By:
1. Title
2. Author
Enter Choice: 1
Enter Search Text: shivan books
```

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main.cpp

```
44 file.close();
45 }
46
47 // Display book details
48 void displayBook() {
49     cout << "Book ID : " << bookID;
50     cout << "Title : " << title;
51     cout << "Author : " << author;
52     cout << "Status : ";
53
54     if (issued)
55         cout << "Issued";
56     else
57         cout << "Available";
58     cout << endl;
59 }
60
61 // Search by title
62 bool searchByTitle(string searchTitle) {
63     return (title == searchTitle);
64 }
65
66 // Search by author
67 bool searchByAuthor(string searchAuthor) {
68     return (author == searchAuthor);
69 }
70
71 // Load book from file
72 bool loadBook(int id) {
73     ifstream file("library.txt");
74     while (file >> bookID) {
75         file.ignore();
76
77         getline(file, title);
78         getline(file, author);
79
80         file >> issued;
81
82         if (bookID == id) {
83             file.close();
84         }
85     }
```

Output

```
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit

Enter Your Choice: 1

Enter Book ID: 56
Enter Book Title: Krishna avadharam
Enter Author Name: Krishna Chaitanya

Book Added Successfully!

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit

Enter Your Choice: 2

===== BOOK LIST =====

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit

Enter Your Choice: 3

Search By:
1. Title
2. Author
Enter Choice: 1
Enter Search Text: shivan books
```

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main.cpp

```
86 return true;
87 }
88
89 file.close();
90 return false;
91 }
92
93 // Issue book
94 void issueBook() {
95     if (issued) {
96         cout << "Book Already Issued!\n";
97     } else {
98         issued = true;
99         updateFile();
100         cout << "Book Issued Successfully!\n";
101     }
102 }
103
104 // Return book
105 void returnBook() {
106     if (!issued) {
107         cout << "Book Was Not Issued!\n";
108     } else {
109         issued = false;
110         updateFile();
111         cout << "Book Returned Successfully!\n";
112     }
113 }
114
115 // Update file
116 void updateFile() {
117     ifstream file("library.txt");
118     ofstream temp("temp.txt");
119
120     int id;
121     string t, a;
122     bool isIssued;
123
124     while (file >> id) {
125         file.ignore();
126
127         getline(file, t);
128         getline(file, a);
129         file >> isIssued;
```

Output

```
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit

Enter Your Choice: 1

Enter Book ID: 56
Enter Book Title: Krishna avadharam
Enter Author Name: Krishna Chaitanya

Book Added Successfully!

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit

Enter Your Choice: 2

===== BOOK LIST =====

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit

Enter Your Choice: 3

Search By:
1. Title
2. Author
Enter Choice: 1
Enter Search Text: shivan books
```

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main.cpp

```
262
263
264 case 2:
265     Book::displayAllBooks();
266     break;
267
268 case 3:
269     Book::searchBook();
270     break;
271
272 case 4:
273     cout << "\nEnter Book ID: ";
274     cin >> id;
275     if (book.findBook(id))
276         book.issueBook();
277     else
278         cout << "\nBook Not Found!\n";
279     break;
280
281 case 5:
282     cout << "\nEnter Book ID: ";
283     cin >> id;
284     if (book.findBook(id))
285         book.returnBook();
286     else
287         cout << "\nBook Not Found!\n";
288     break;
289
290 case 6:
291     cout << "\nThank You!\n";
292     return 0;
293
294 default:
295     cout << "\nInvalid Choice!\n";
296 }
297
298 return 0;
299 }
```

Output

```
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit

Enter Your Choice: 1

Enter Book ID: 56
Enter Book Title: krishna avadharam
Enter Author Name: Krishna Chaitanya

Book Added Successfully!

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit

Enter Your Choice: 2

===== BOOK LIST =====

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit

Enter Your Choice: 3

Search By:
1. Title
2. Author
Enter Choice: 1
Enter Search Text: shivan books
```

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Search

ENG IN

19:41

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